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$$\begin{aligned}\lim_{x \rightarrow 0} \left(\frac{\ln(1 - 3x)}{\tan 5x} \right) &= \frac{0}{0} \\ \stackrel{(H)}{=} \lim_{x \rightarrow 0} \left(\frac{-\frac{3}{1 - 3x}}{5 \sec^2(5x)} \right) \\ &= \lim_{x \rightarrow 0} \left(-\frac{3}{(1 - 3x) \cdot 5 \sec^2(5x)} \right) \\ &= -\frac{3}{(1 - 3 \cdot 0) \cdot 5 \sec^2(5 \cdot 0)} \\ &= -\frac{3}{5}\end{aligned}$$

References